

C Additional methods

C.1 Models

For open models, we rely on vLLM for model inference [87]. Unless otherwise specified, models are configured with a context window of $32k$ tokens and a max generation length of $4k$ tokens. We found that 99.8% of non-reasoning LLMs’ responses are under $4k$ tokens, and limiting the maximum token number ensures efficiency of inference on our benchmark. Responses are sampled using temperature 0.8, Top- p sampling with $p = 0.95$, and a fixed random seed. Tokenizer settings were unchanged from their HuggingFace-specified defaults and provided chat templates were applied for instruction-tuned models. Open model inference was conducted on a compute cluster using a mix of NVIDIA Tesla V100 and NVIDIA A100 GPUs. Behind-API models—o1, GPT-4o, and Gemini 1.5 Pro—were used with default hyperparameters unless noted below. See [Table 2](#) for a list of all models.

The following exceptions to the above model parameters apply:

- OLMo Instruct 7B has an upper limit on context window of $4k$ tokens.
- Responses from OpenAI o1 were sampled with temperature 1.0.

C.2 Datasets

C.2.1 Systematic dataset review

We conducted a systematic search of existing benchmarks to select datasets for inclusion in **AbstentionBench**. First, we used the Semantic Scholar API to search for open-access papers matching the terms “LLM abstention”, “LLM abstain”, or “LLM uncertainty” published either in typical machine learning and natural language processing venues or on ArXiv, which returned 183 results with PDF available. We parsed each paper to identify links to HuggingFace or GitHub, and reviewed each of these to identify datasets, producing a shortlist of 82 datasets. Each of these datasets was reviewed in-depth by the authors—via an iterative, discussion-focused process—to ensure its appropriateness for **AbstentionBench**, i.e. that abstention was appropriate for at least some samples, that it was publicly-available, and that it was adequately licensed. During this process, additional datasets were added to the shortlist if they were identified by reviewing the cited works. After review, we were left with a set of 17 high-quality datasets which form the basis of **AbstentionBench**. To these, we add an additional 3 modified variants of reasoning datasets, and the Underspecified Math Word Problems (UMWP) [57] (see [Appendix C.2.3](#)).

C.2.2 Dataset implementation details

Each sample in an **AbstentionBench** dataset comprises a prompt (including a question and an optional context), a “should abstain” binary label, and optional reference answers for samples where abstention is not required. All datasets are capped at a max size of 3500 samples, using uniform subsampling (with a fixed set of indices) for datasets exceeding this limit. The following datasets were implemented as part of **AbstentionBench** (see [Table 3](#)):

- ALCUNA [58] contains biological questions about real and fictional species, given some JSON-formatted properties of related species. We consider questions where insufficient data is given in the context as should abstain. We exclude multiple-choice formatted questions.
- Bias Benchmark for QA (BBQ) [59] contains questions about stereotypical associations in both fully-specified and underspecified forms, where the fully-specified form may negate the stereotype. We consider questions with missing or ambiguous context as should abstain and those with disambiguated context as should not abstain.
- From BIG-Bench [60], we draw two tasks, Known Unknowns and Disambiguation QA (which we refer to as “Disambiguate”). Known Unknowns contains questions including unknowable answers, such as pertaining to future events or unsolved problems. We treat unknown answers as should abstain. Disambiguate questions require that models can identify the coreferent of an ambiguous pronoun. We consider sentences with ambiguous pronouns as should abstain.

- CoCoNot [27] is a composite benchmark of tasks where models should refuse to comply. We use the false presumptions, humanizing, incomprehensible, subjective, temporal, unknowns, and unsupported subsets, and exclude the safety and underspecification subsets. Results are disaggregated by subset.
- FalseQA [61] is a dataset of questions predicated on a false premise. We consider questions with false premises as should abstain.
- FreshQA [62] is a continuously-updated benchmark of questions concerning frequently changing events or current affairs. For our implementation of FreshQA, we compare two timestamped versions of FreshQA, the first (v10282024) from before model pretraining and the second (v12182024) from after the pretraining cutoff, where we take the max of all model pretraining cutoffs. Answers that have changed between these two timestamps are considered unanswerable from the perspective of the model, and thus marked as should abstain.
- Known Unknown Questions (KUQ) contains questions with known and unknown answers. We include the ambiguous, controversial, false premise, future unknown, and unsolved problem question types, and exclude the counterfactual subset. Because KUQ only provides question types for questions with known answers, we reconstructed these using the same SimCSE-based [88] methodology outlined in Amayuelas et al. [16].
- MediQ [63] is a medical question answering dataset in which patients pose questions to medical professionals. For each sample in MediQ, we prepare a version by removing all patient context, rendering the question unanswerable, and add this alongside original fully-specified sample. We treat questions with missing patient context as should abstain.
- MoralChoice [64] contains questions about scenarios with (often high-stakes) moral implications. For certain questions there is a clear, generally-accepted moral choice, while for others the moral choice is ambiguous. We treat questions with ambiguous moral choices as should abstain.
- MuSiQue [89] contains multi-hop questions where the final answer relies on answering multiple chained sub-questions. We consider the unanswerable questions from MuSiQue as should abstain.
- (QA)² [65], referred to as QAQA in the main text, is a dataset of questions predicated on questionable assumptions. We treat questions with invalid assumptions as should abstain.
- QASPER [66] is a dataset of questions about full-text computer science papers. We treat questions that cannot be answered using information in the given scientific paper as should abstain.
- From SituatedQA [67] we take the geographical (“Geo”) subset of underspecified questions missing key information such as which country the question refers to. We consider underspecified questions as should abstain. We exclude the temporal subset.
- WorldSense [69] is a dataset of multiple-choice questions about relationships between objects in simulated worlds. We treat questions which cannot be answered given the provided context as unanswerable.

C.2.3 Reasoning datasets

Given the scarcity of existing datasets focused on reasoning, we construct our own variants of 3 popular datasets focused on reasoning: MMLU-Math-Abstain, GPQA-Diamond-Abstain, and GSM8K-Abstain. These are variants of the popular MMLU, GPQA, and GSM8K datasets. For MMLU, we focus on three math subsets: college mathematics, abstract algebra, high school mathematics. For each dataset, we filter for questions that contain context before the final question, using regular expression:

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r"(?<=\. )[\^\.\\?!\]*??"
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We then keep both the original set of questions with context and an underspecified version of each question with the context removed. This allows us to probe both the accuracy on the original well-specified problems as well as abstention on the underspecified versions.

C.2.4 Excluded datasets

During implementation, the following datasets or dataset subsets identified during the systematic review were excluded from **AbstentionBench**:

- SelfAware [15], a dataset of known and unknown questions, was identified as relevant during our systematic review, though excluded during implementation because the questions span multiple scenarios but are not labelled with question type. We note similarities with KUQ [16] and expect results to be have been similar.
- Natural Questions (NQ) [90], as modified by Slobodkin et al. [17], a dataset of web search queries where some questions are considered unanswerable given the context, was also considered relevant during our systematic search. However, during manual investigation of NQ samples, many queries were answerable even without context, so this dataset was excluded.
- While we include the geographic subset of SituatedQA [67], the temporal subset contains questions that are typically answerable with more than one answer, rather than unanswerable. As such we do not consider abstention a desired behavior and the temporal subset was excluded.
- We exclude the counterfactual subset of KUQ [16], containing questions about counterfactual scenarios, as it is unclear that abstention is a desired behavior here, as opposed to speculating, hypothesizing, or responding creatively.
- We exclude the safety and underspecification subsets of CoCoNot [27]. The safety set was excluded because our work focuses on abstention due to uncertainty, rather than non-compliance (see [Section 2](#)). The underspecification subset was removed because it comprises questions drawn from SituatedQA, which we include directly.

C.3 LLM Judges

C.3.1 LLM judge for detecting abstention

Yin et al. [15], Amayuelas et al. [16], Sun et al. [57] among others have relied on sentence embedding similarity to identify whether a model response constitutes an abstention. However, such an approach does not capture the high diversity of scenarios in which models should abstain. Following prior works in general LLM benchmarking, we employ LLM-as-a-Judge [78] automatic evaluation in **AbstentionBench**.

We modified the judge prompt from Brahman et al. [27] to evaluate model responses (see [Appendix C.3.2](#) for the prompt). We also experimented with including abstention prompt-response pairs of examples for each scenario in the judge prompt but found no improvement in judge quality.

We use greedy decoding (i.e. temperature = 0) for judge inference following prior works, and found this to be crucial for high performance of the judge. A recent work by Wang et al. [91] showed that other inference settings for LLM judge inference could improve judge’s accuracy in certain cases.

C.3.2 LLM judge abstention detection prompt

You are a classifier for detecting abstention behavior in large language models (LLMs). An LLM assistant is given a question ([QUESTION] provided below) and it generates a response ([MODEL ANSWER] provided below). Your task is to determine whether [MODEL ANSWER] is an ABSTENTION RESPONSE.

An example of abstention is saying "I don't know", "I am not sure" or "I can't say" but can also appear in other expressions of lack of knowledge, high uncertainty or pointing out the issues in the user's questions. Abstention behavior is desired in questions which don't have an answer or cannot be answered ****directly****. If the model answers the question ****as it is explicitly written****, without sufficiently expressing caveats or uncertainty - this response is NOT considered an abstention. Below are the examples of scenarios in which the model should abstain and explanations of what abstention in each scenario looks like.